

PharmacoEconomics

Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach --Manuscript Draft--

Manuscript Number:		
Full Title:	Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach	
Article Type:	Original Research Article	
Funding Information:	Horizon 2020 (754923)	Dr Annette T. Byrne
Abstract:	Objectives: Bevacizumab was approved for first-line treatment of metastatic colorectal cancer (mCRC) in 2004. However, adding bevacizumab to treatment consistently fails to be cost-effective due to modest response rates. Recently, the European Commission (EC) funded ANGIOPREDICT consortium (www.angiopredict.com) identified a link between bevacizumab treatment response and intermediate-to-high chromosomal instability (CIN) in mCRC. Thus, the objective of the current study was to compare the cost-effectiveness of adding bevacizumab to first-line chemotherapy in the bevacizumab responsive CIN subtype across three European countries (Germany, Ireland and Spain) with varying costs of care and reimbursement policies. Methods: We developed an open-source health economic model to estimate cost-effectiveness. The ANGIOPREDICT cohort informed progression risks and cause-specific mortality. Health utilities and adverse events probabilities were obtained from the literature. Costs were derived from surveys of collaborating consortium hospitals in Germany, Ireland and Spain that participated in the recently completed EC funded COLOSSUS translational study (ANGIOPREDICT successor initiative) and the literature. Sensitivity analyses included individual and simultaneous variation of input parameters from a priori defined distributions. Results: Adding bevacizumab was only cost-effective when willingness to pay (WTP) thresholds were multiple times larger than currently accepted cut-offs. The highest incremental cost-effectiveness ratio was in Germany at €241,188 per quality-adjusted life year (QALY), while the lowest was in Ireland at €180,477 per QALY. All deterministic and probabilistic sensitivity analyses demonstrated that these results were robust. Conclusions: Even for patients with mCRC manifesting improved outcomes, adding bevacizumab to first-line chemotherapy is invariably not cost-effective in any of the countries examined. Variability in pricing, healthcare costs and willingness to pay thresholds across countries did not commute this result.	
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Author Comments:	Editor-in-Chiefs, PharmacoEconomics, Re: Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach Dear Editors-in-Chief, On behalf of the European Commission funded Colossus consortium (www.colossusproject.eu), I am writing to submit our manuscript entitled "Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach" for consideration for publication in PharmacoEconomics. With almost 2 million new cases diagnosed annually worldwide, colorectal cancer is the second leading cause of global cancer deaths. Metastatic colorectal cancer (mCRC) has a particularly low survival rate, just 14% at five years. Targeted therapies exist in mCRC to improve response rates and extend survival. As the first clinically applied angiogenesis inhibitor in any disease, bevacizumab may be the most successful of such targeted therapies. Nonetheless, bevacizumab is typically not considered a cost-effective addition to the chemotherapeutic regimen, as the degree of clinical benefit is limited. However, this paradigm may be shifting; emerging evidence suggests there exists a subtype of patients with metastatic colorectal cancer (mCRC) for whom the effect of bevacizumab is beyond what has previously been shown in mCRC. We have identified chromosomal instability (CIN) as a novel predictive biomarker of BVZ response in mCRC (Smeets et al. Nat Comms 2018 9;4119, van Dijk et al. J Clin Oncol 2018 36 (20):2052-2060). In this context, mCRC patients belonging to intermediate-to-high CIN subtypes manifest improved outcomes following bevacizumab combination therapy. In the currently submitted manuscript our analysis therefore uniquely focuses on whether individuals with specific genomic cancer subtypes may derive a more cost-effective benefit from existing treatment. Overall, we seek to study the complex interplay between clinical efficacy and cost-effectiveness. Specifically, our study evaluates the cost-effectiveness of adding bevacizumab to first-line chemotherapy for mCRC patients with intermediate-to-high chromosomal instability (CIN) across high, middle, and low health-expenditure European countries. To our knowledge, this is the first cost-effectiveness analysis of this widely-used therapy in biomarker-defined subgroups. We believe it addresses a critical gap in the literature by incorporating a precision medicine approach to mCRC treatment, offering insights that extend beyond traditional cost-effectiveness evaluations. By studying the economic implications of precision medicine across healthcare systems spanning the spectrum of expenditure within the EU, our findings can significantly contribute to the ongoing discourse on the economic evaluation of cancer therapies to help inform future healthcare resource allocation decisions. Moreover, we have developed an open-source model with transparent reporting

	(https://github.com/JonathanBriody/COLOSSUS_Model) to align with the principles of open science, fostering reproducibility and adaptation to other settings to enhance the impact of this work by encouraging further research and innovation in the health economics of cancer care. The data, models, and methodologies used in this research are not proprietary and we make these freely available online with our manuscript for examination and re-analysis. This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 754923. The material presented and views expressed here are the responsibility of the authors only. The publication of study results was not contingent on the sponsor's approval or censorship of the manuscript. This manuscript is original work that has not been published or submitted for publication elsewhere. All authors have reviewed and approved the manuscript and agreed to its submission to PharmacoEconomics. This manuscript presents an open-source, reproducible modeling framework that clarifies the pharmacoeconomic trade-offs of targeted oncology therapies—especially their high acquisition costs—and evaluates strategies to improve cost-effectiveness through precision medicine. The work aligns with PharmacoEconomics' mission to advance the clear communication of complex pharmacoeconomic issues related to all healthcare interventions. Thank you for considering our submission; we look forward to your review. Sincerely, Jonathan Briody, Health Economist School of Population Health RCSI University of Medicine and Health Sciences
Suggested Reviewers:	Daniel A. Goldstein danielg3@tauex.tau.ac.il Medical oncologist and health-economist who has published extensively on the cost-effectiveness of cancer drugs, including multiple studies on bevacizumab; strong fit for a price/value analysis in targeted oncology. Peter S. Hall p.s.hall@ed.ac.uk Clinician-economist leading cancer health-economics research in the UK; experienced with economic evaluations of oncology therapies and methods relevant to HTA decision-making. Turgay Ayer ayer@isye.gatech.edu Operations-research modeler with deep expertise in individual-level simulation and cost-effectiveness modeling in oncology; strong on transparent, reproducible modeling. Kelvin K. W. Chan kelvin.chan@sunnybrook.ca Medical oncologist with numerous CEAs and real-world value studies of targeted cancer therapies; familiar with payer perspectives and comparative value across jurisdictions. Laura Vallejo-Torres laura.vallejo@ulpgc.es Spanish health economist who led empirical work on Spain's cost-effectiveness thresholds; ideal for evaluating Spanish WTP choices and methodological justification. Afschin Gandjour a.gandjour@fs.de German health economist with publications on AMNOG, willingness-to-pay, and oncology evaluations; well placed to appraise the German pricing/HTA angle. Sabine Vogler

	sabine.vogler@goeg.at European expert on pharmaceutical pricing and cross-country price comparisons; brings essential context for your international price scenarios. Brian Godman brian.godman@strath.ac.uk Medicines-policy scholar focused on pricing, access, and biosimilars in oncology; relevant for considerations around bevacizumab biosimilars and budget impact. Maarten J. IJzerman maarten.ijzerman@unimelb.edu.au Health-economics leader in precision medicine and companion diagnostics; well suited to evaluate biomarker-guided treatment modeling and implementation. Sebastian Stintzing sebastian.stintzing@charite.de German medical oncologist with major colorectal cancer trial work comparing anti-EGFR agents vs bevacizumab; brings clinical depth and German-system insight.
Additional Information:	
Question	Response
What is the word count of the manuscript, excluding references?	3,757
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Has this article been previously submitted to another journal and rejected (either before or after peer review)? If yes, please make a note here and then provide all editorial/referee comments, along with an explanation of how these comments have been addressed, in a covering letter as part of your submission.	Yes, before peer review, and it was then internally transferred to this journal, there were no comments to address.
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Please copy here the key points (2-3 sentences summarizing, in non-technical language, the key findings/implications of the manuscript) that you have included after the Abstract in your manuscript. Please note, key points are not required for Letters to the Editor, Editorials or Commentaries.	Key Points Targeted oncology therapies exist to improve response rates and extend survival, but are frequently not cost-effective when their degree of clinical benefit is considered in the context of their cost. We assess whether a precision medicine approach to a traditionally cost-ineffective therapy for metastatic colorectal cancer translates into improvements in cost-effectiveness in healthcare systems spanning the spectrum of EU expenditure. Model source code, usage instructions, and study data are open-source, fostering adaptation to other settings. Infrastructure decision-makers can freely use to model intervention in real-world patients helps inform future allocation decisions – providing a resource to improve public health while promoting innovation in the health economics of cancer care.

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02/09/2025

Editor-in-Chiefs, *PharmacoEconomics*,

Re: Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability:
Cost-Effectiveness Analysis for a Novel Precision Treatment Approach

Dear Editors-in-Chief,

On behalf of the European Commission funded Colossus consortium (www.colossusproject.eu), I am writing to submit our manuscript entitled **"Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach"** for consideration for publication in *PharmacoEconomics*.

With almost 2 million new cases diagnosed annually worldwide, colorectal cancer is the second leading cause of global cancer deaths. Metastatic colorectal cancer (mCRC) has a particularly low survival rate, just 14% at five years. Targeted therapies exist in mCRC to improve response rates and extend survival. As the first clinically applied angiogenesis inhibitor in any disease, bevacizumab may be the most successful of such targeted therapies. Nonetheless, bevacizumab is typically not considered a cost-effective addition to the chemotherapeutic regimen, as the degree of clinical benefit is limited. However, this paradigm may be shifting; emerging evidence suggests there exists a subtype of patients with metastatic colorectal cancer (mCRC) for whom the effect of bevacizumab is beyond what has previously been shown in mCRC. We have identified chromosomal instability (CIN) as a novel predictive biomarker of BVZ response in mCRC (Smeets et al. *Nat Comms* 2018 9;4119, van Dijk et al. *J Clin Oncol* 2018 36 (20):2052-2060). In this context, mCRC patients belonging to intermediate-to-high CIN subtypes manifest improved outcomes following bevacizumab combination therapy.

In the currently submitted manuscript our analysis therefore uniquely focuses on whether individuals with specific genomic cancer subtypes may derive a more cost-effective benefit from existing treatment. Overall, we seek to study the complex interplay between clinical efficacy and cost-effectiveness. Specifically, our study evaluates the cost-effectiveness of adding bevacizumab to first-line chemotherapy for mCRC patients with intermediate-to-high chromosomal instability (CIN) across high, middle, and low health-expenditure European countries. To our knowledge, this is the first cost-effectiveness analysis of this widely-used therapy in biomarker-defined subgroups. We believe it addresses a critical gap in the literature by incorporating a precision medicine approach to mCRC treatment, offering insights that extend beyond traditional cost-effectiveness evaluations.

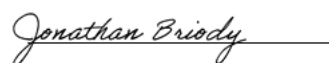
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This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 754923. The material presented and views expressed here are the responsibility of the authors only. The publication of study results was not contingent on the sponsor's approval or censorship of the manuscript. This manuscript is original work that has not been published or submitted for publication elsewhere.

All authors have reviewed and approved the manuscript and agreed to its submission to PharmacoEconomics.

This manuscript presents an open-source, reproducible modeling framework that clarifies the pharmacoeconomic trade-offs of targeted oncology therapies—especially their high acquisition costs—and evaluates strategies to improve cost-effectiveness through precision medicine. The work aligns with PharmacoEconomics' mission to advance the clear communication of complex pharmacoeconomic issues related to all healthcare interventions. Thank you for considering our submission; we look forward to your review.

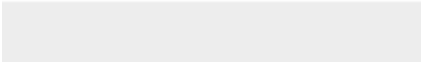
Sincerely,



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Running Title: Cost-Effectiveness of Bevacizumab as Precision Medicine for Metastatic Colorectal Cancer.

Abstract

Objectives: Bevacizumab was approved for first-line treatment of metastatic colorectal cancer (mCRC) in 2004. However, adding bevacizumab to treatment consistently fails to be cost-effective due to modest response rates. Recently, the European Commission (EC) funded ANGIOPREDICT consortium (www.angiopredict.com) identified a link between bevacizumab treatment response and intermediate-to-high chromosomal instability (CIN) in mCRC. Thus, the objective of the current study was to compare the cost-effectiveness of adding bevacizumab to first-line chemotherapy in the bevacizumab responsive CIN subtype across three European countries (Germany, Ireland and Spain) with varying costs of care and reimbursement policies.

Methods: We developed an open-source health economic model to estimate cost-effectiveness. The ANGIOPREDICT cohort informed progression risks and cause-specific mortality. Health utilities and adverse events probabilities were obtained from the literature. Costs were derived from surveys of collaborating consortium hospitals in Germany, Ireland and Spain that participated in the recently completed EC funded COLOSSUS translational study (ANGIOPREDICT successor initiative) and the literature. Sensitivity analyses included individual and simultaneous variation of input parameters from a priori defined distributions.

Results: Adding bevacizumab was only cost-effective when willingness to pay (WTP) thresholds were multiple times larger than currently accepted cut-offs. The highest incremental cost-effectiveness ratio was in Germany at €241,188 per quality-adjusted life year (QALY), while the

lowest was in Ireland at €180,477 per QALY. All deterministic and probabilistic sensitivity analyses demonstrated that these results were robust.

Conclusions: Even for patients with mCRC manifesting improved outcomes, adding bevacizumab to first-line chemotherapy is invariably not cost-effective in any of the countries examined. Variability in pricing, healthcare costs and willingness to pay thresholds across countries did not commute this result.

Key words:

Colorectal neoplasms, cost-effectiveness analysis, Quality-adjusted life years, Markov model, mCRC, Open-Source Software

JEL CLASSIFICATION CODES: I19, I1, I, I11, I1, I,

Key Points

Targeted oncology therapies exist to improve response rates and extend survival, but are frequently not cost-effective when their degree of clinical benefit is considered in the context of their cost.

We assess whether a precision medicine approach to a traditionally cost-ineffective therapy for metastatic colorectal cancer translates into improvements in cost-effectiveness in healthcare systems spanning the spectrum of EU expenditure.

Model source code, usage instructions, and study data are open-source, fostering adaptation to other settings. Infrastructure decision-makers can freely use to model intervention in real-world patients helps inform future allocation decisions – providing a resource to improve public health while promoting innovation in the health economics of cancer care.

1. Introduction

With almost 2 million new cases diagnosed annually worldwide, colorectal cancer is the third most commonly diagnosed cancer in men, the second most commonly diagnosed in women [1], and the second leading cause of cancer deaths worldwide [2]. Following advances in primary and adjuvant treatments, survival time continues to improve for colorectal cancer patients, with an average five-year net survival rate for colon cancer patients in EU countries of 59% [3]. However, this is not reflected in metastatic colorectal cancer (mCRC), in which patients have a dramatically lower 5-year survival rate of just 14% [4].

Targeted therapy for cancer directly inhibits cell proliferation, differentiation, and migration [5]. Similarly, the micro-environment of the tumour, including the local blood vessels and immune cells, may be modified in order to disrupt growth and to improve immune surveillance and response [6]. Approved for the first-line treatment of patients with mCRC in combination with chemotherapy by the U.S. Food and Drug Administration (FDA) in 2004 and the European Medicines Agency (EMA) in 2005, as the first clinically applied angiogenesis inhibitor in any disease, bevacizumab may be the most successful of all targeted therapies [7].

Although widely used as an addition to the first-line chemotherapeutic regimen, bevacizumab is typically not considered cost-effective in the US, Canada, Asia or Europe [8–14]. This may correspond to a limited overall benefit of bevacizumab in the clinic [15,16]. However, there is emerging evidence that for a subtype of patients with mCRC, the effect of bevacizumab is beyond what has previously been shown in mCRC. Chromosomal instability (CIN) is a form of genomic instability in solid tumours resulting from defects in mitosis and pre-mitotic replication stress [17]. The ANGIOPREDICT consortium investigated responsiveness to bevacizumab following addition

to chemotherapy in relation to degree of CIN, identifying a group of patients with mCRC (manifesting intermediate-to-high CIN) with improved outcomes [18–20].

Patients with mCRC bearing tumours with intermediate-to-high chromosomal instability (CIN) demonstrated significantly improved progression-free survival (PFS) and overall survival (OS) following addition of bevacizumab to chemotherapy compared to chemotherapy alone (Online Resource Supplementary Fig. S1 and S2) [19]. Here, we hypothesised that improvements in effectiveness within the CIN intermediate/high subtype might translate into improvements in the cost-effectiveness of bevacizumab treatment for mCRC patients with such CIN, providing greater support for clinical use [21]. Moreover, we have recently developed a prototype liquid biopsy for this biomarker (Manuscript Accepted) which may in the future be clinically applied to select for those patients with mCRC who will not respond to combination bevacizumab therapy.

Consequently, the aim of the present health economics study was to conduct the first cost-effectiveness analysis of combination bevacizumab therapy for the emergent predictive CIN subtype.

2. Description of Patients and Treatment Regimens

Our cost-effectiveness study is a secondary data analysis of an existing publication comparing the addition of bevacizumab to chemotherapy treatment [19]. Thus, this work did not involve additional patients. In Smeets et al [19] the ANGIOPREDICT consortium retrospectively collected tumour biopsies and clinical data from patients with mCRC. Our study includes data from these patients with advanced (locally irresectable or metastatic) colorectal cancer (CRC) who began treatment with chemotherapy and bevacizumab between July 2004 and April 2012 [19]. Tumour tissue from patients with mCRC fulfilling these criteria was gathered from the tissue biobanks of

the Royal College of Surgeons in Ireland (RCSI) Beaumont Hospital, Ireland (n = 29), the University of Heidelberg (UHEI), Germany (n = 107) and VU University Medical Center (VUMC), Netherlands (n = 34). The analysis also used data on mCRC tumours from the CAIRO and CAIRO II trials [19]. These open-label, multicentre randomised phase 3 trials, recruited adult patients at centres in the Netherlands with histologically proven, previously untreated mCRC not amenable to curative surgery. Patients were treated with chemotherapy alone without bevacizumab, as described previously [23,24]. In ANGIOPREDICT patients received a fluoropyrimidine (FP) chemotherapy backbone in combination with either irinotecan (IRI) or oxaliplatin (OX), in CAIRO patients received FP chemotherapy alone, or with the addition of IRI. The sex breakdown between the ANGIOPREDICT cohort and CAIRO was comparable (61% male ANGIOPREDICT, 62% male CAIRO), as was the age (42% >65 in ANGIOPREDICT, 44% > 65 in CAIRO) T-classification (64% classified as 3 in ANGIOPREDICT, 69% classified as 3 in CAIRO) and N-classification (34% as 2 in ANGIOPREDICT, 32% as 2 in CAIRO).

Pathologists assessed tissue samples, and clinical data (including grading, staging, treatment follow-up, age and sex) was gathered from patients' records. Samples were stratified into patients with mCRC who received bevacizumab combined with chemotherapy (185 ANGIOPREDICT collected tumour biopsies), and patients with mCRC who received chemotherapy alone (19 ANGIOPREDICT collected tumour biopsies and 205 tumours collected from CAIRO phase 3 trial patients [19]. PFS and OS were chosen as clinical endpoints. PFS was defined as the time to progressive disease or any cause of mortality from beginning chemotherapy treatment with bevacizumab in the experimental strategy arm and without bevacizumab in the standard of care arm. Patients who withdrew from bevacizumab therapy for reasons other than progression or death were censored on date of withdrawal. OS was defined as the time beginning therapy in the standard

of care/experimental strategy arm to any cause of mortality. Patient data were administratively censored after 60 months. In multivariable Cox regression, Smeets et al [19] revealed that patients with intermediate-to-high chromosomal instability exhibited an improved response to bevacizumab and chemotherapy relative to receiving chemotherapy alone. Bevacizumab significantly improved progression-free survival (HR = 0.68; 95% CI 0.53--0.87) and overall survival (HR = 0.65; 95% CI 0.49--0.86). The analysis was corrected for various potential confounders, and the relationship between chromosomal instability and response to bevacizumab remained unchanged.

Consequently, in this study, we considered two lines of treatment with specific utilities and costs in each line. The current first-line standard of care for mCRC in the countries studied is 5-FU, leucovorin, and oxaliplatin (FOLFOX), given with or without bevacizumab. Patients who progress on this therapy typically receive second-line treatment with 5-FU, leucovorin, and irinotecan (FOLFIRI) [19,25,26]. However, FOLFOX and FOLFIRI can be used interchangeably in the first- and second-line care sequence because phase III trials comparing the two regimens have demonstrated that both provide similar response rates and PFS [27]. Our state-transition model evaluated two different treatment strategies. The strategy of interest was first-line treatment with FOLFOX combined with bevacizumab, and second-line treatment with FOLFIRI on progression. The comparator was first-line treatment with FOLFOX without bevacizumab, then FOLFIRI on progression. After initiating the second-line treatment, patients subsequently experienced progression until death [28].

3. Materials and Methods

Model Structure

In the present study we sought to establish the cost-effectiveness of adding bevacizumab to first-line chemotherapy in this new mCRC subtype. The challenge to drawing compelling and generalisable conclusions from such an analysis is the fact that it is not possible to simply extrapolate cost-effectiveness data globally. Variability in pricing, healthcare costs, WTP, and multiple international payer perspectives are such that cost-effectiveness results may be expected to differ across countries. In response to this challenge, we leverage an existing mCRC focused consortium ‘COLOSSUS’ (<https://www.colossusproject.eu/>) to access clinical expertise in modelling the data across the COLOSSUS consortium countries of Germany, Ireland, and Spain. COLOSSUS is an EU-funded H2020 programme seeking to advance cost-effective mCRC care [29]. Current health expenditure in Germany is the largest share of gross domestic product (GDP) of all 27 EU Member States [30], while health spending in Spain is below the EU average [31]. Ireland's public health spending is in the middle, historically low but increasing dramatically over the past half-century [32]. By choosing these countries as representative cases, we aim to capture the diversity of health expenditure patterns across the EU to produce conclusions on cost-effectiveness that are applicable to member states with high, medium and low public health investment.

We applied a decision-analytic framework in mCRC developed as part of the COLOSSUS project to model the cost-effectiveness of the addition of bevacizumab to FOLFOX first-line chemotherapy versus chemotherapy alone. We constructed a Markov model to calculate the lifetime costs and health outcomes of each treatment strategy in mCRC. The intention-to-treat (ITT) population we modelled in this study is patients with mCRC in Germany, Ireland and Spain

from a healthcare payer perspective according to the guidelines for health economic evaluations in these countries [33]. Effectiveness is measured in quality-adjusted life years (QALYS). Per national guidelines on economic evaluations in Germany, Ireland, and Spain a discount rate of 3%, 4% and 3% per annum respectively was applied to cost input and health output parameters in the countries studied [34]. All model inputs are described in Table 1. The primary outcomes of the model were the total costs, QALYs, and the incremental cost-effectiveness ratio (ICER). Prior cost-effectiveness analysis in mCRC has demonstrated that the probability of transitions, adverse events, and quality of life under treatment may be assumed to be generalisable across countries [35]. Consequently, shared values were applied to all countries under study in the interest of concision. Country specific differences in cost were included.

An open-source health-economic model was created and is freely available online [36]. The model simulated the health state transitions following a diagnosis of mCRC. A cycle length of 14 days was chosen representing the time between the administrations of chemotherapy doses. A 5-and-a-half-year time horizon was also chosen; this was appropriate to estimate life expectancy in this patient population [28]. As shown in Figure 1, we established the following health states in the model: (1) first-line treatment prior to progression, (2) second-line treatment following progression, and (3) death. The model was analysed in R software version 4.1.2 (<http://www.r-project.org>), and the 2022 Consolidated Health Economic Evaluation Reporting Standards (CHEERS) guidelines were applied in this study, as per field standards in a health economic analysis, to describe evaluation and results [37].

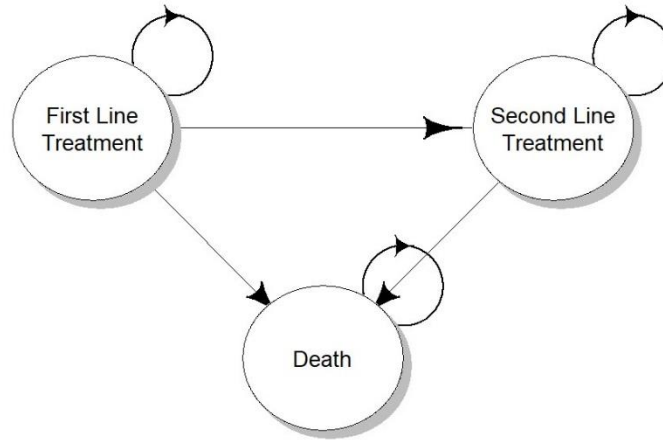


Fig. 1 Three-state Markov model for mCRC

Progression Risk, Adverse Events and Mortality:

We used the clinical data informing the Smeets et al [19] study to estimate the clinical course for patients treated under standard of care and the hazard ratio results published in this study to estimate outcomes for patients receiving the addition of bevacizumab as the experimental strategy. For the purpose of model validation, we replicate the results found in Smeets et al [19] in our own study (Online Resource Supplementary Fig. S1 and S2) and demonstrate that the patient transitions generated by our Markov model match closely with those PFS and OS curves generated by Smeets et al [19] (Online Resource Supplementary Fig. S3).

The data informing the Smeets et al [19] survival curves were used to fit parametric survival models. To produce estimates of progression and mortality matching those reported in this publication, parametric models were fit to exponential, Weibull, Gompertz, generalised gamma, log-normal and log-logistic survival models, respectively. Weibull models have been demonstrated to be more suitable for modelling events occurring in mCRC [38], and provided the best fit for PFS and OS per visual inspection (Online Resource Supplementary Fig. S4) and

goodness-of-fit statistics (Online Resource Supplementary Table S1). Hence, we estimated the transition probabilities for each cycle based on the Weibull distribution model.

For standard of care, the transition from first to second-line treatment in each cycle was calculated from the Smeets et al [19] PFS data; OS data informed transitions from first-line treatment to death. The hazard ratios published in the Smeets et al [19] study were then applied to adjust transition probabilities given the addition of bevacizumab to chemotherapy. The transition probability of second-line treatment into death is as informed by Wen et al [39]. The frequency of adverse events was as described in the TREE trial of FOLFOX with or without bevacizumab, which reflects the occurrence of such events in treated mCRC [40]. As Smeets et al [19] did not demonstrate an influence of chromosomal instability on the incidence of treatment-related adverse events in patients with mCRC, we did not differentiate the risk of adverse events from the general patient population.

Utility Estimates

The quality of life utilities of a cycle spent in each of the model health states have been estimated to be 0.85 for the first-line health state and 0.65 for the second-line health state in the published literature on patients with mCRC [41]. While some more recent studies exist, the Ramsey et al [41] analysis is one of a limited number to apply several independent quality-of-life measures to a large sample of actual CRC patients to capture health state-specific utilities. The authors employ rigorous modelling techniques to derive precise utility estimates. Accordingly, the reported utility values are within the range reported for mCRC in a recent systematic review of 57 studies of CRC [42], align with questionnaire responses from patients in mCRC clinical trials [43] and continue to be applied in contemporary studies of mCRC [44,45].

Adverse events (AEs) can reduce patients' utility. Most commonly in mCRC this includes grade 3/4 leukopenia, diarrhoea, and nausea [46]. As weighted by the incidence reported in the literature, duration-adjusted AE disutility was subtracted from the baseline utility to determine the total utility of each health state [10,47,48]. The disutility and duration of such events were derived from the literature [49].

Cost Estimates

Country-specific cost data associated with each health state were gathered via a direct survey of COLOSSUS consortium hospitals in the countries studied and the previous literature [50–56]. All cost estimates were adjusted to 2024 Euro values, costs derived from the literature were inflation-adjusted as necessary using appropriate consumer price indices for health, per the Online Resource (Supplementary Information [SI]). From a payer perspective, direct healthcare costs were considered. Costs of both first-line and second-line treatments were included. As informed by previous studies, the primary cost inputs chosen included drug regimen costs, administration costs, and treatment-related adverse event costs per patient, per cycle [35,59]. The probability of each AE, and its corresponding duration, were used to calculate the overall effect on costs associated with each treatment. Interventions are considered cost-effective if the ICER is below the thresholds healthcare systems are willing to pay per QALY. The commonly accepted WTP threshold of efficiency is €45,000 per QALY gained in Ireland and €30,000 per QALY gained in Spain [55,60]. There are a lack of formal German thresholds in cost-effectiveness analyses [61]. To contextualise our analysis, a threshold of €78,871 is applied as suggested by a recent assessment of the German monetary value of health, conducted to support cost-effectiveness analysis in this country [62].

4. Sensitivity Analysis

A number of additional analyses were conducted to further support the robustness of the model results to variations in parameters or assumptions. We applied deterministic methods (point estimates with an appropriate range) and probabilistic methods (parameterised distributions) to conduct a sensitivity analysis of our results (as described in detail in the Supplementary Material). The baseline values, ranges, and distributions used for model parameters in these analyses are presented in Table 1.

Table 1 Model Parameters Values: Baseline, Ranges and Distributions for Sensitivity Analysis

Parameter	Base Case Value	Minimum Value	Maximum Value	Source	Distribution
Cost					
(Per Cycle)					
FOLFOX	ESP:	ESP:	ESP:	ESP:	ESP:
	323.39	258.71	388.07	[54]	γ A,B(100, 3.23)
	DE:	DE:	DE:		DE:
	656.31	525.04	787.57		γ A,B(100, 6.56)
	IRE:	IRE:	IRE:		IRE:
	157.77	126.22	189.32		γ A,B(100, 1.58)
FOLFIRI	ESP:	ESP:	ESP:	ESP:	ESP:
	147.31	117.85	176.77	[55]	γ A,B(100, 1.47)

	DE:	DE:	DE:		DE:
	673.26	538.61	807.91		γ A,B(100, 6.73)
	IRE:	IRE:	IRE:		IRE:
	167.10	133.68	200.52		γ A,B(100, 1.67)
Bevacizumab	ESP:	ESP:	ESP:	ESP:	ESP:
	1399.33	1119.46	1679.20	[55]	γ A,B(100,13.99)
	DE:	DE:	DE:		DE:
	1517.69	1214.15	1821.23		γ A,B(100, 15.18)
	IRE:	IRE:	IRE:		IRE:
	1425.84	1140.67	1711.01		γ A,B(100, 14.26)
Administration	ESP:	ESP:	ESP:	ESP:	ESP:
Cost	332.39	265.91	398.87	[55]	γ A,B(100, 3.32)
	DE:	DE:	DE:	DE:	DE:
	1927.72	1542.18	2313.26	[51]	γ A,B(100, 19.28)
	IRE:	IRE:	IRE:		IRE:
	406.50	325.20	487.80		γ A,B(100, 4.07)
Subtyping	One Off			Cost Calculation for the	
Test	Cost:	320.00	480.00	Subtyping Test	γ A,B(100, 4.00)
	400			Described in the	

Supplementary
Material.

Adverse Event Cost					
Leukopenia	ESP:	ESP:	ESP:	ESP:	ESP:
	5710.97	4568.78	6853.16	[52]	γ A,B(100, 57.11)
	DE:	DE:	DE:	DE:	DE:
	4188.31	3350.65	5025.97	[50]	γ A,B(100, 41.88)
	IRE:	IRE:	IRE:		IRE:
	3485.23	2788.18	4182.28		γ A,B(100,34.85)
Diarrhea	ESP:	ESP:	ESP:	ESP:	ESP:
	305.55	244.44	366.66	[53]	γ A,B(100, 3.06)
	DE:	DE:	DE:	DE:	DE:
	1982.68	1586.14	2379.22	[50]	γ A,B(100, 19.83)
	IRE:	IRE:	IRE:		IRE:
	1792.82	1434.26	2151.38		γ A,B(100, 17.93)
Nausea	ESP:	ESP:	ESP:	ESP:	ESP:
	35.11	28.09	42.13	[53]	γ A,B(100, 0.35)

DE:	DE:	DE:	DE:	DE:
1307.72	1046.18	1569.26	[56]	γ A,B(100, 13.08)
IRE:	IRE:	IRE:	IRE:	IRE:
502.69	402.15	603.23		γ A,B(100, 5.03)

**Adverse
Event
Incidence -
With
bevacizumab**

Leukopenia	0.033	0.027	0.040	[40]	β A,B(96.64, 2805.43)
Diarrhea	0.053	0.042	0.064	[40]	β A,B(94.65, 1693.63)
Nausea	0.033	0.027	0.040	[40]	β A,B(96.64, 2805.43)

**Adverse
Event
Incidence -
Without
bevacizumab**

Leukopenia	0.019	0.015	0.023	[40]	β A,B(98.09, 5100.34)
Diarrhea	0.159	0.127	0.191	[40]	β A,B(83.94, 443.99)
Nausea	0.159	0.127	0.191	[40]	β A,B(83.94, 443.99)
Adverse Event Incidence – Second Line Treatment					
Leukopenia	0.009	0.008	0.011	[40]	β A,B(99.05, 10456.81)
Diarrhea	0.131	0.105	0.157	[40]	β A,B(86.76, 575.07)
Nausea	0.120	0.096	0.144	[40]	β A,B(87.86, 643.03)
Utility (Per Cycle)					
First Line Treatment	0.85	0.68	1	[41]	β A,B(15.41, 2.72)

Second Line Treatment	0.65	0.52	0.78	[41]	β A,B(32.96, 17.75)
Adverse Event Disutility					
Leukopenia	0.45	0.36	0.54	[49]	β A,B(54.55, 66.67)
Diarrhea	0.19	0.15	0.23	[49]	β A,B(80.81, 344.51)
Vomiting	0.36	0.29	0.43	[49]	β A,B(63.64, 113.14)
Hazard Ratios					
First to Second Line with Bevacizumab	0.68	0.53	0.87	[19]	lnorm()
First Line to Dead with Bevacizumab	0.65	0.49	0.86	[19]	lnorm()
Probabilities					

Death under Second-Line Treatment	0.17	0.12	0.22	[39]	β A,B(36.69, 179.13)
Discount Rate					
Costs:	0.03/0.04/0.0	0.00	0.06/0.08/0.0	[34]	Uniform()
Germany/Ireland/Spain	3		6		
Outcomes:	0.03/0.04/0.0	0.00	0.06/0.08/0.0	[34]	Uniform()
Germany/Ireland/Spain	3		6		

***Abbreviations:** *ESP:* Spanish, *DE:* German, *IRE:* Irish, costs. **Distribution:** Applied in sensitivity analysis.

Source: Where data is drawn from the literature, the source is described.

5. Results

The ICERs of bevacizumab, in addition to standard therapy, exceed conventional cost-effectiveness thresholds in all three countries considered. While we estimate the same effects in each case, costs - and consequently ICERs - differ significantly. The highest ICER was in Germany at €241,188 per QALY, and the lowest was in Ireland at €180,477 per QALY. In Spain, the ICER was €201,346 per QALY. While the reported ICERs exceed thresholds by a considerable margin, they are in accordance with other cost-effectiveness analyses of bevacizumab in the U.S., U.K., Australia, Israel, and Canada [35]. Notwithstanding, results are even less cost-effective in these studies which assess patients without chromosomal instability. All values are summarised in Table 2.

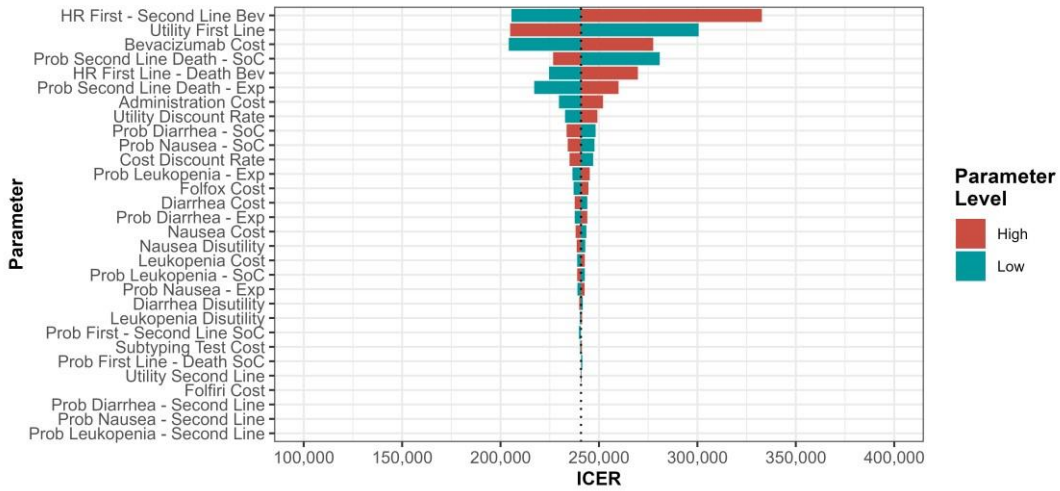
Table 2. Cost-Effectiveness Results for the Addition of Bevacizumab to
First-line mCRC Chemotherapy in ANGIOPREDICT

Result	Germany	Ireland	Spain
Total incremental cost (Euro)	41,699	30,847	34,811
Total incremental effect (QALYs)	0.173	0.171	0.173
ICER (Euro/QALY)	241,188	180,477	201,346

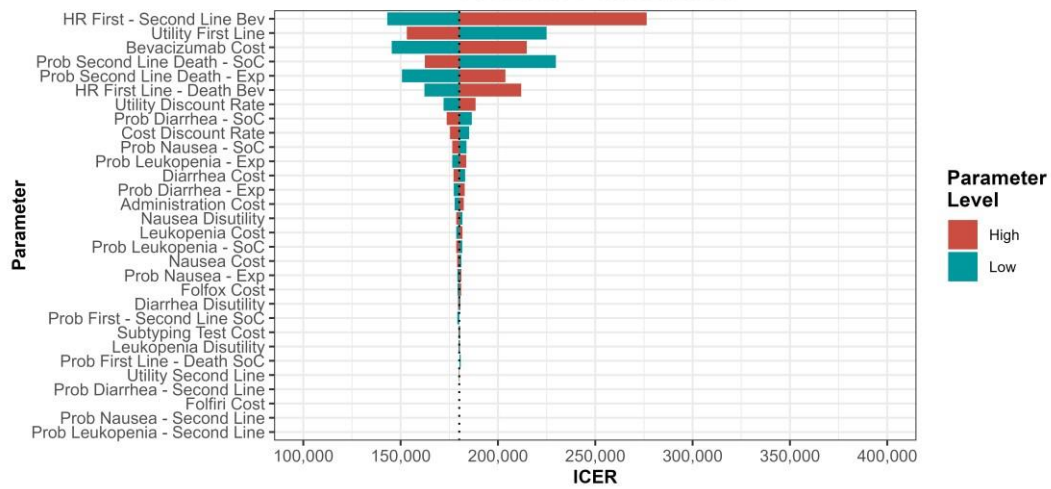
*ICER = incremental cost-effectiveness ratio; QALY = quality-adjusted
life year*

In Figure 2, we evaluated the relative weight of each parameter individually in our model on the overall uncertainty of cost-effectiveness in a one-way sensitivity analysis. Across the univariate analyses, parameters that influenced ICERs were comparable between countries. The most significant influence on the ICER consistently came from the hazard ratio describing bevacizumab's ability to stop disease progression, the level of utility in this first-line health state and the cost of bevacizumab. Within a relatively narrow range, there was a limited influence of variation in other parameters. However, results remained robust to adjustments, with an ICER consistently above the threshold of WTP per QALY in the three countries studied.

Tornado Plot Germany



Tornado Plot Ireland



Tornado Plot Spain

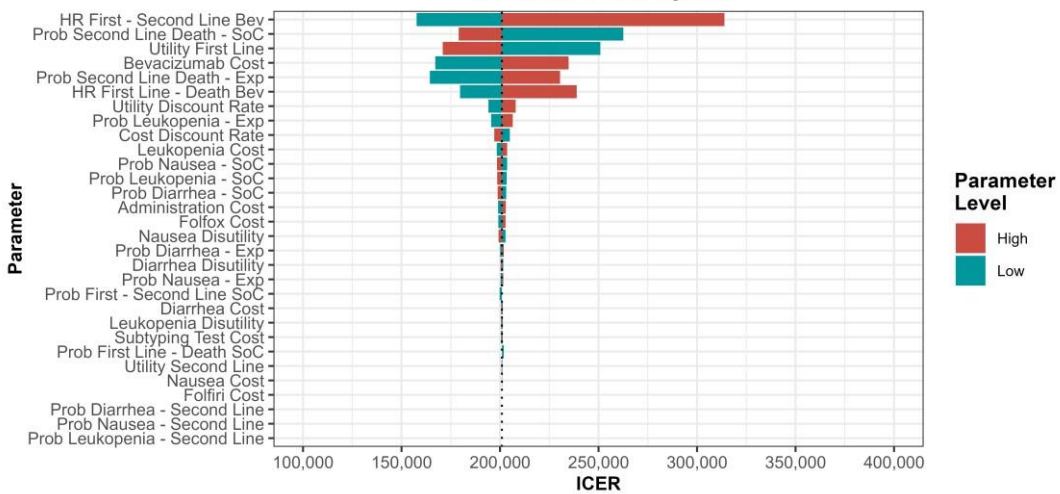
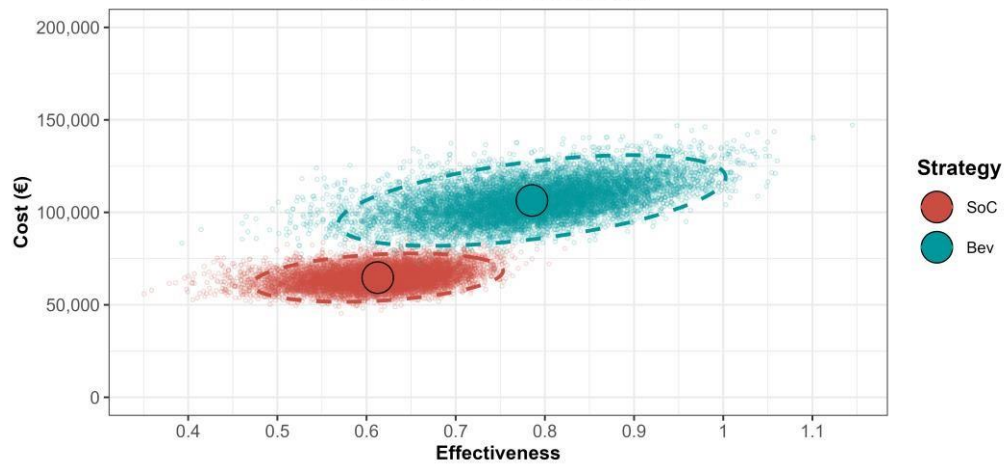


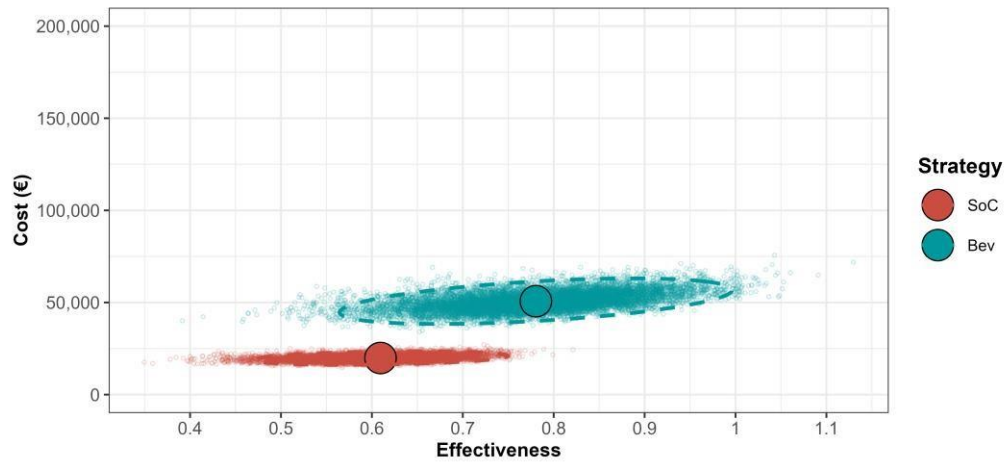
Fig. 2 Tornado Plot of Germany, Ireland and Spain, respectively. In this diagram, we denote parameter levels as high or low to correspond to parameter values above or below the median point estimates reported in Table 1. This highlights where differences in the magnitude of input values (across the minimum and maximum values described in Table 1) result in changes in the ICER value

We also conducted a probabilistic sensitivity analysis (PSA) to determine the consequences of parameter uncertainty in our model for cost-effectiveness outcomes. The results of the PSA illustrated in the cost-effectiveness scatter plot in Figure 3 demonstrate a greater degree of uncertainty surrounding the mean point estimates for the addition of bevacizumab to the standard of care, illustrated by the magnitude of the 95%-confidence ellipses surrounding these estimates. However, no ICER value is sufficiently close to the WTP threshold to be considered cost-effective, regardless of the country under study. Cost-effectiveness acceptability curves (CEACs) are derived from the PSA, as per Figure 4, to demonstrate that for the addition of bevacizumab to be considered cost-effective in the countries studied, the WTP thresholds would have to increase to many times their current value, often three to six times currently accepted levels.

Scatter Plot Germany



Scatter Plot Ireland



Scatter Plot Spain

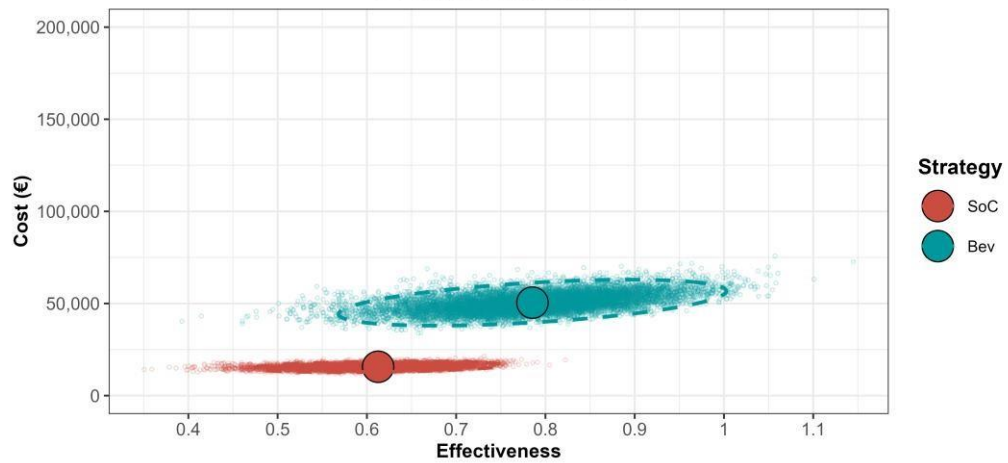
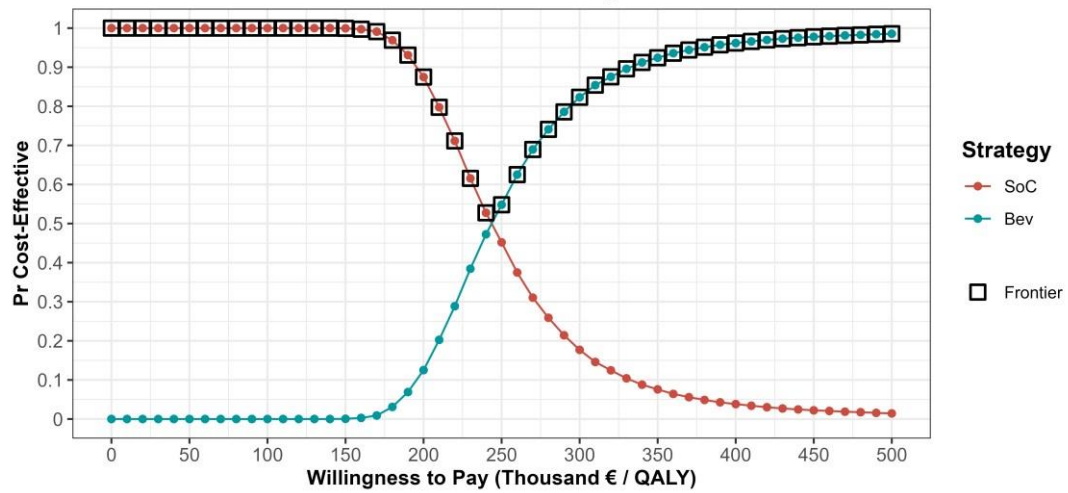
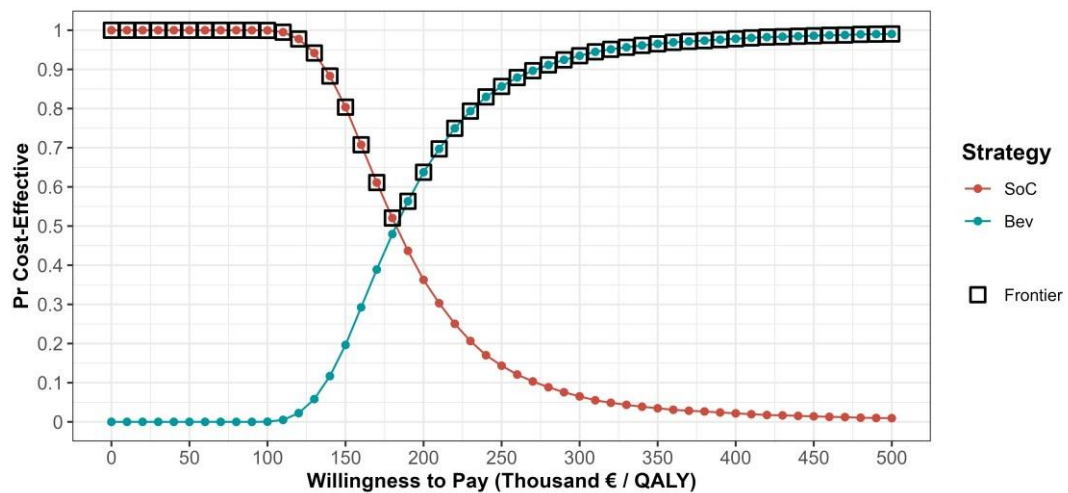


Fig. 3 Cost-effectiveness Scatter Plot of Germany, Ireland and Spain, respectively. In total, 10,000 replications were completed to determine the possible distribution of the resulting costs and effects with and without bevacizumab. The ellipses around points represent 95% confidence incidences

CEAC Plot Germany



CEAC Plot Ireland



CEAC Plot Spain

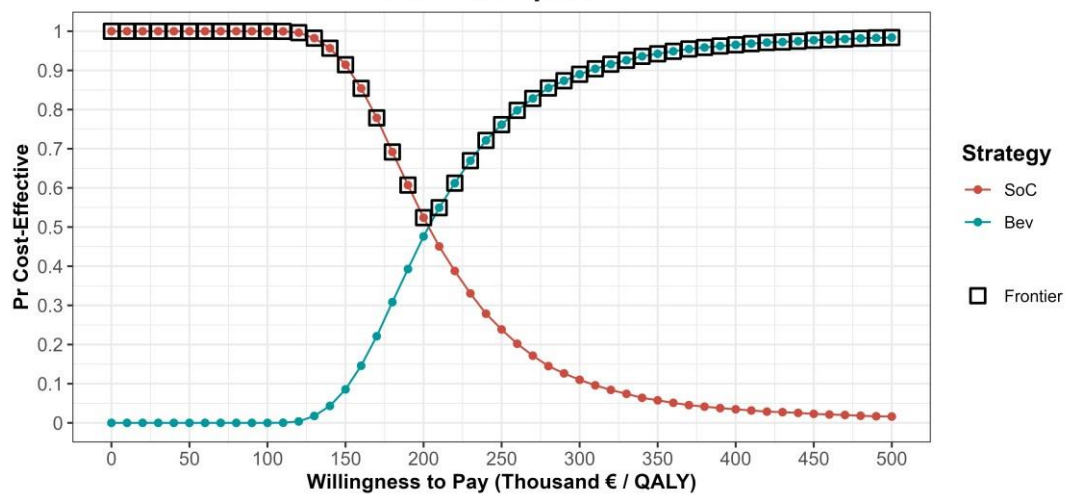


Fig. 4 Cost-Effectiveness Acceptability Curves (CEACs) of Germany, Ireland and Spain, respectively

6. Discussion

For novel interventions in cancer to constitute a useful addition to care they must be cost-effective, that is, gains from introducing a therapy must outweigh those foregone when reallocating existing health budgets. Bevacizumab has consistently failed to demonstrate cost-effectiveness, largely due to its significant cost [8,12,14]. Our study tests whether improvements in the effectiveness of bevacizumab for mCRC patients with chromosomal instability may be enough to offset this cost, providing a subtype with greater support for clinical use. PFS and OS for the Smeets et al [19] intermediate-to-high CIN subtype are approximately one and three months greater than previously observed in the literature on the addition of bevacizumab to FOLFOX [28,63]. This additional benefit to bevacizumab is particularly significant, given that PFS and OS improvements of less than two months have formerly been deemed sufficient for regulatory approval in adding this drug to FOLFOX therapy [64]. We hypothesised that this improved efficacy may consequently increase efficiency, such that bevacizumab may finally reach a reasonable cost-per-QALY threshold to be deemed cost-effective.

However, according to our cost-effectiveness analysis, treatment with bevacizumab continues to provide minimal incremental benefit at a high incremental cost per QALY, across the three European countries examined. The magnitude of the effect of cost of bevacizumab observed in the univariate analysis on this conclusion is expected given that bevacizumab is on average three to ten times larger than any other cost included in our analysis. The significance of the bevacizumab hazard ratio and utility in the first-line health state is also not surprising, given that the mechanism

of action in bevacizumab is to keep patients in the first-line health state for as long as possible. Variability in the PSA is similarly likely to be a consequence of the leverage bevacizumab has in affecting the model outcome, both in terms of cost and effectiveness.

The price of drug therapies in a country is determined by a process of negotiation between the reimbursement authority in that country and the drug manufacturer. Such authorities set the maximum level of payment their country's national health insurance system will make for a medication. Hofmarcher et al [65] establish that the absolute cost of cancer care is consequently lower in lower-wealth countries, similar to our findings, they find that Germany is among the highest-cost countries for care, Spain is among the lowest, and Ireland is midway. This is significant because, per international protocols, the provision of cancer therapies is largely the same across these countries [66]. Here, we establish the cost-effectiveness of bevacizumab in mCRC in the COLOSSUS consortium countries of Germany, Ireland and Spain, countries which span the spectrum of health expenditure within the EU and within which we could directly engage clinical experts as part of the COLOSSUS consortium. Our results are consistent across variability in pricing, healthcare costs, willingness to pay thresholds and multiple international payers – the improved clinical benefit of bevacizumab in the CIN high subtype does not offset the additional cost of the biological treatment.

One limitation of the present study, is that we assess cost-effectiveness for an originator drug at a time of increased approval of biosimilar therapeutics. Biosimilar provision is particularly likely to grow in Europe when one considers that biological treatments account for almost a third of all pharmaceutical spending [67]. The uptake of an appropriately discounted bevacizumab biosimilar option may therefore provide an alternative that would make our analysis cost-effective. That is,

perhaps health systems could reduce their expenditure while maintaining a therapy that sustains the safety and efficacy of bevacizumab.

Therapeutic options with equivalent clinical profiles to that of the originator have been approved by the FDA and the European Commission, including Vegzelma (Celltrion), Zirabev (Pfizer), MVASI (Amgen) and Aybintio (MSD) [68]. However, the actual uptake of these drugs remains low in the three countries studied [67]. Additionally, the reductions in price for these biosimilars are typically a maximum of 30% off the originator [69]. Repeating our analysis with a 60% reduction in the price of Bevacizumab does not result in cost-effectiveness in Germany, Ireland or Spain (analysis not reported here for brevity). Thus, we believe that the introduction of biosimilars as currently described would not generate sufficient cost savings in the health systems under study to alter the conclusions of our analysis.

In Germany, the practical value of our analysis for decision makers needs to be clarified. The German statutory authority that assesses the degree of benefit of emerging therapies, the Federal Joint Committee (GBA), applies an assortment of methodologies for assessing cost-effectiveness when making reimbursement decisions [70]. These methods may not incorporate those techniques applied in this analysis. However, by providing detailed information on input parameters and reporting ICERs directly, we contribute data that supports any such analysis conducted by the GBA.

Finally, we developed our model applying data from the Smeets et al [19] study, subsequently, it may not exhaustively reflect the real-world experience of a population of patients. However, the ranges we apply across various sensitivity analyses will likely account for variations between populations. Our sensitivity analysis will similarly account for any possible variations in true

utility values as a product of applying utility data in our study from an external source, per common practice in health economic analysis [10]. Variations in cost are similarly subsumed in the ranges and distributions applied across sensitivity analyses.

7. Conclusions

Patients with mCRC belonging to intermediate-to-high CIN subtypes manifest improved outcomes following bevacizumab combination therapy [19]. However, without analogous improvements in efficiency we cannot conclude that the expected benefits of care outweigh the expected costs in Germany, Ireland or Spain. Demonstrating consistent results in such countries with disparate health expenditure profiles suggest it is unlikely that analysis of other EU member states would provide evidence for cost-effectiveness.

8. Data Availability Statement

The source code for the open-source cost-effectiveness model developed for this study, usage instructions and study data are freely available under a Creative Commons license via DOI: DOI removed for blinded review purposes, or through the following public repository: url removed for blinded review purposes

9. References

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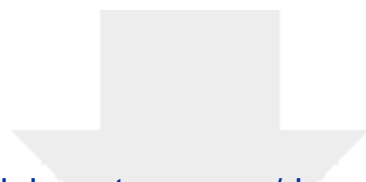
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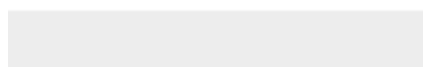
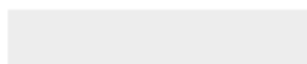
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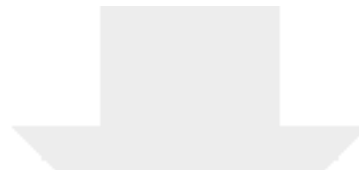


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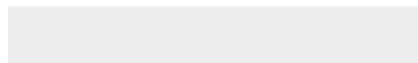
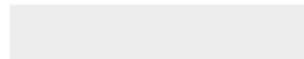
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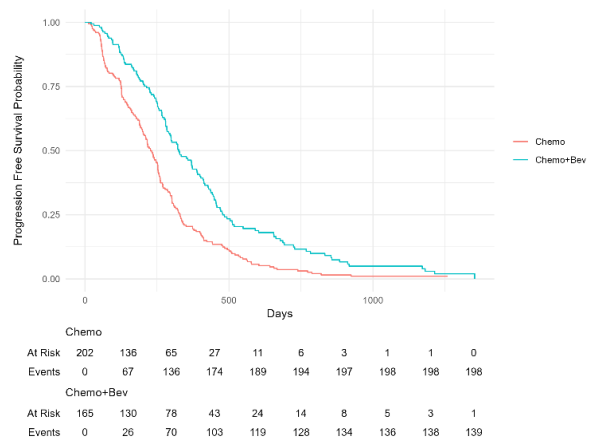
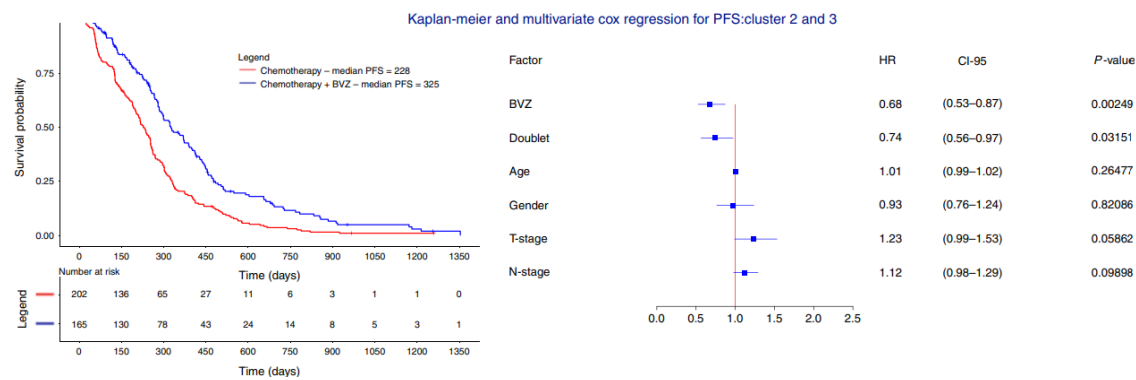


Figure S1: PFS published in Smeets et al. (2018), followed by our replication of their results. The first part of the figure is reprinted with permission from Smeets, D., Miller, I. S., O'Connor, D. P., Das, S., Moran, B., Boeckx, B., ... & Byrne, A. T. (2018). Copy number load predicts outcome of metastatic colorectal cancer patients receiving bevacizumab combination therapy. *Nature Communications*, 9(1), 1–16. The second part shows our replication results.

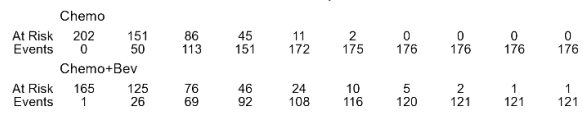


Figure S2: OS published in Smeets et al. (2018), followed by our replication of their results. The first part of the figure is reprinted with permission from Smeets, D., Miller, I. S., O'Connor, D. P., Das, S., Moran, B., Boeckx, B., ... & Byrne, A. T. (2018). Copy number load predicts outcome of metastatic colorectal cancer patients receiving bevacizumab combination therapy. *Nature Communications*, Electronic Supplementary Material. The second part shows our replication results.

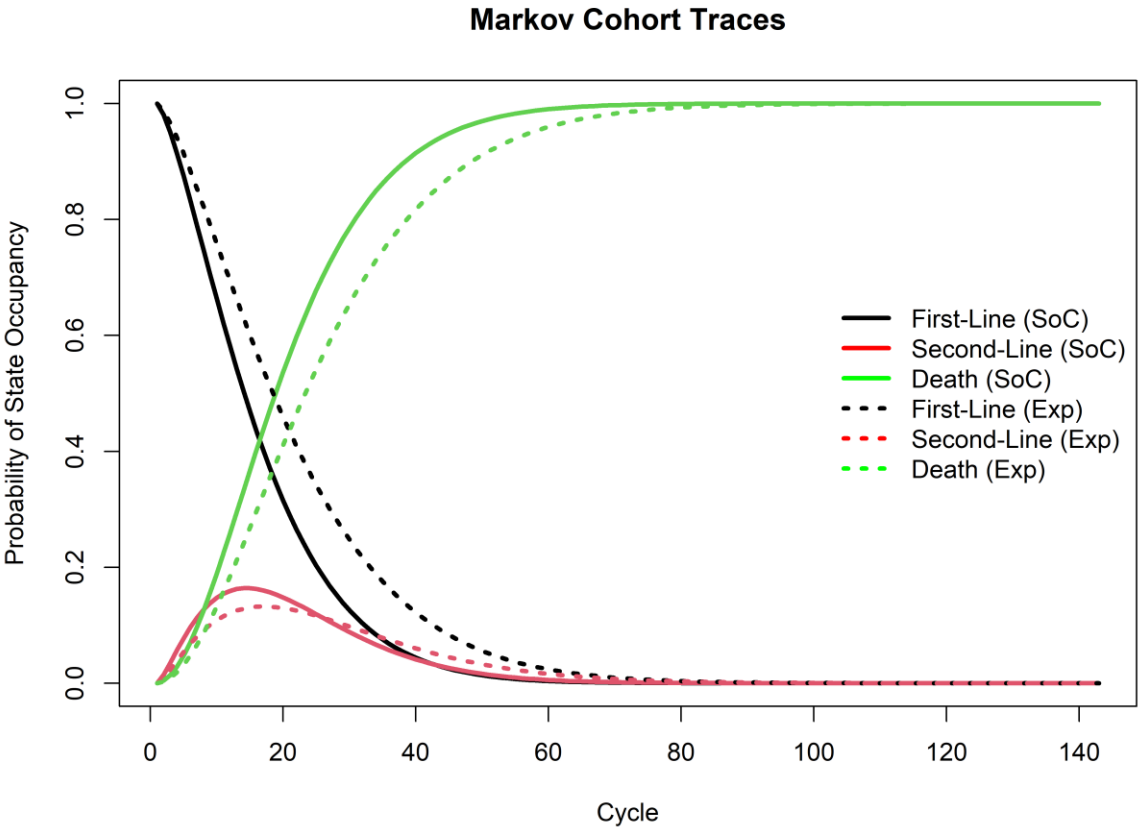


Figure S3: Cohort Traces generated by the Markov model simulation.

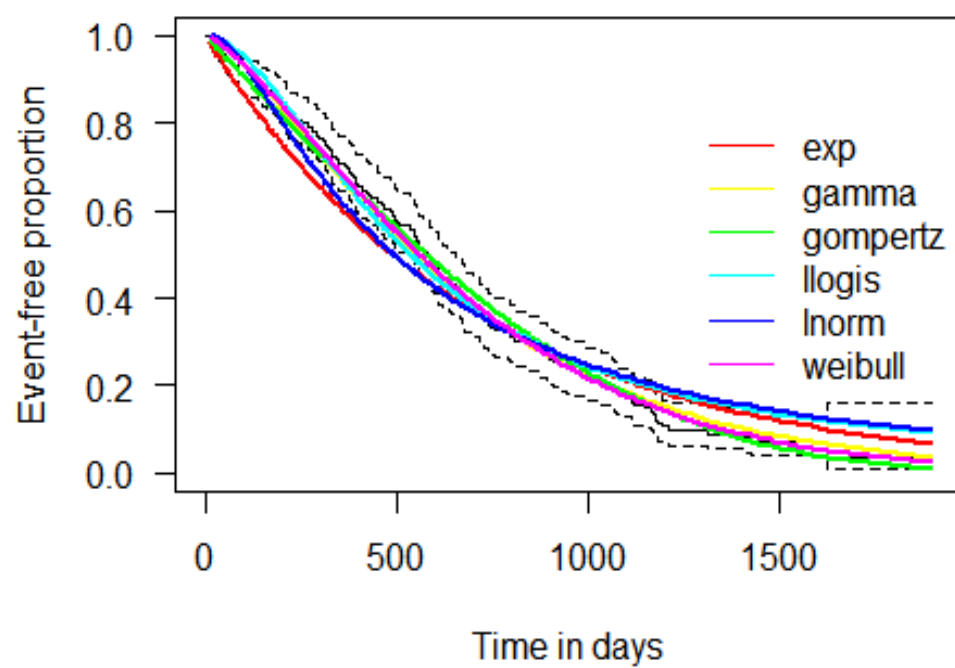
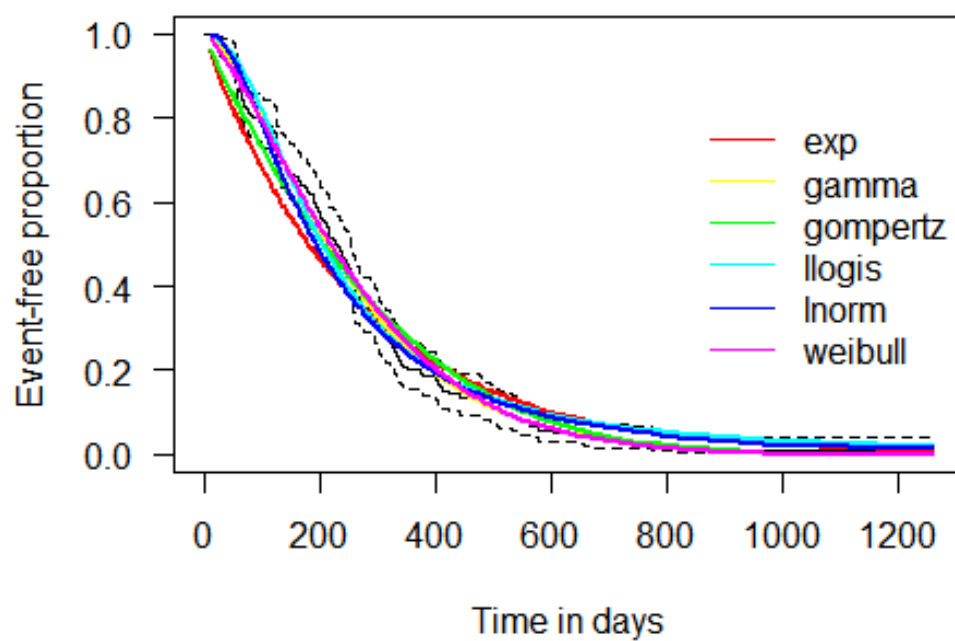


Figure S4: Visual inspection of the relative conformity of parametric models to the original PFS and OS Kaplan-Meier survival curves.

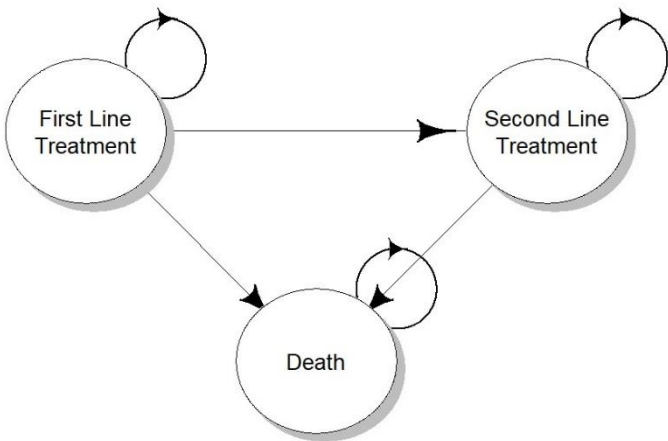


Figure 1: Three-state Markov model for mCRC

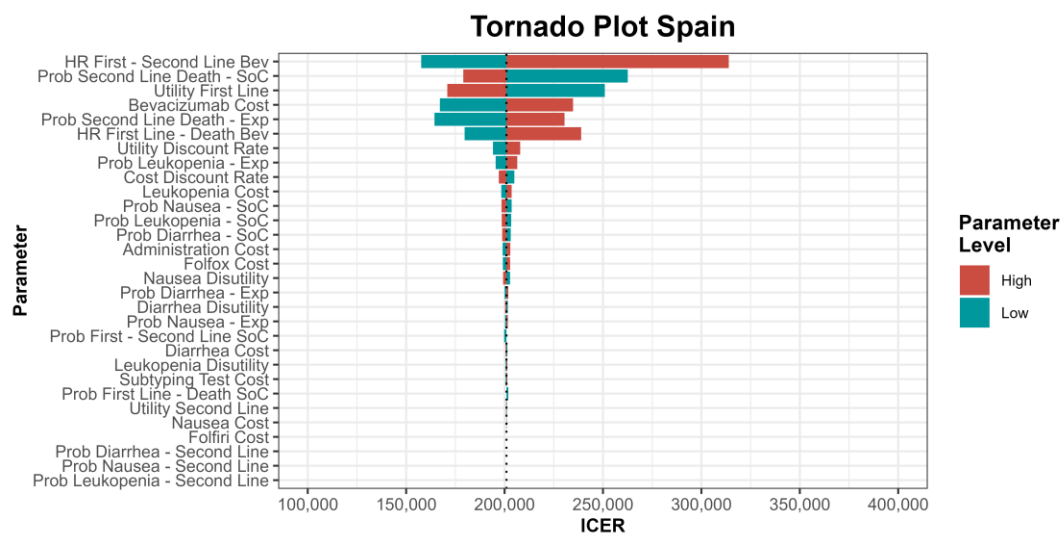
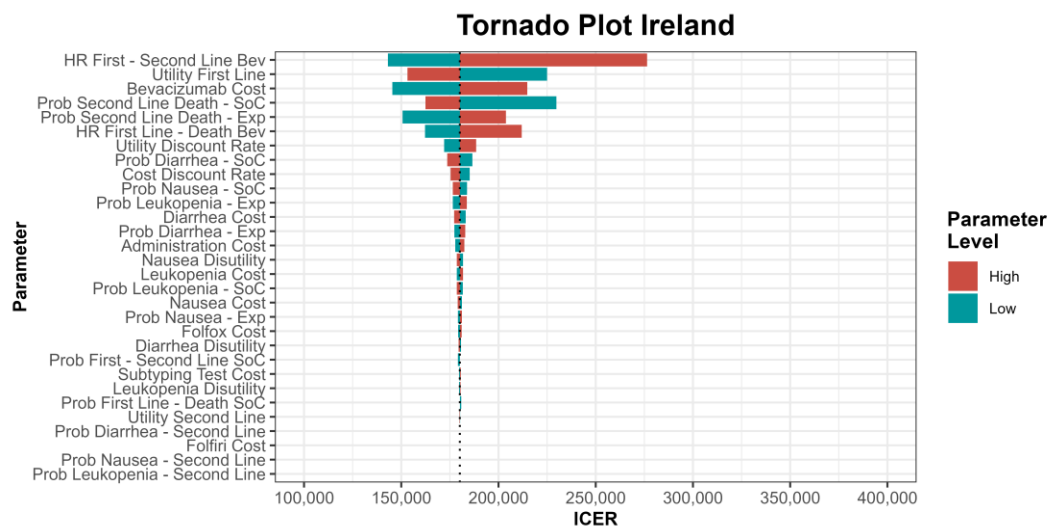
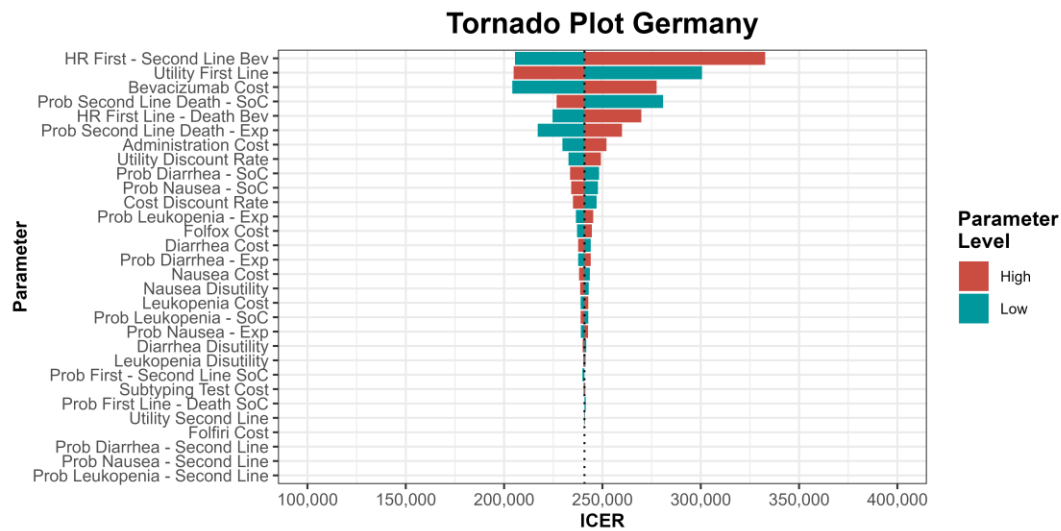


Figure 2: Tornado Plot of Germany, Ireland and Spain, respectively. In this diagram, we denote parameter levels as high or low to correspond to parameter values above or below the median point estimates reported in Table 1. This highlights where differences in the magnitude of input values (across the minimum and maximum values described in Table 1) result in changes in the ICER value.

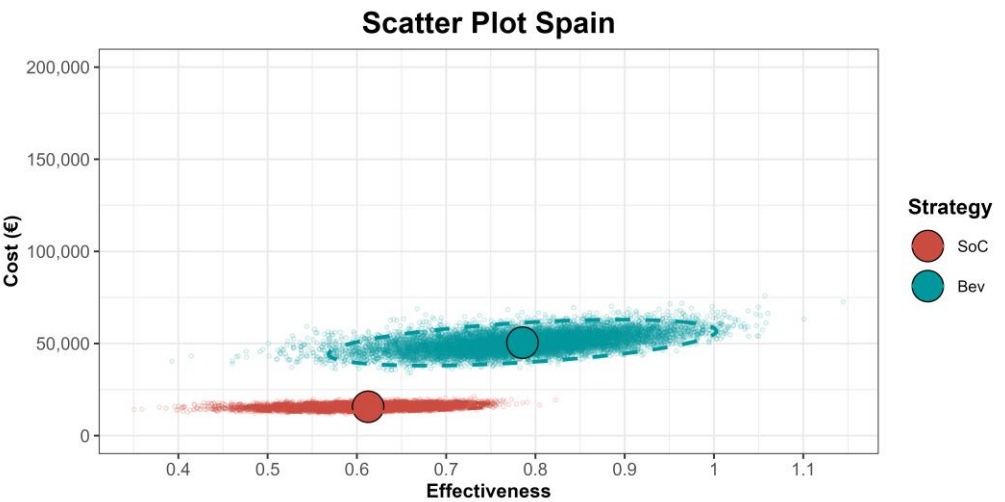
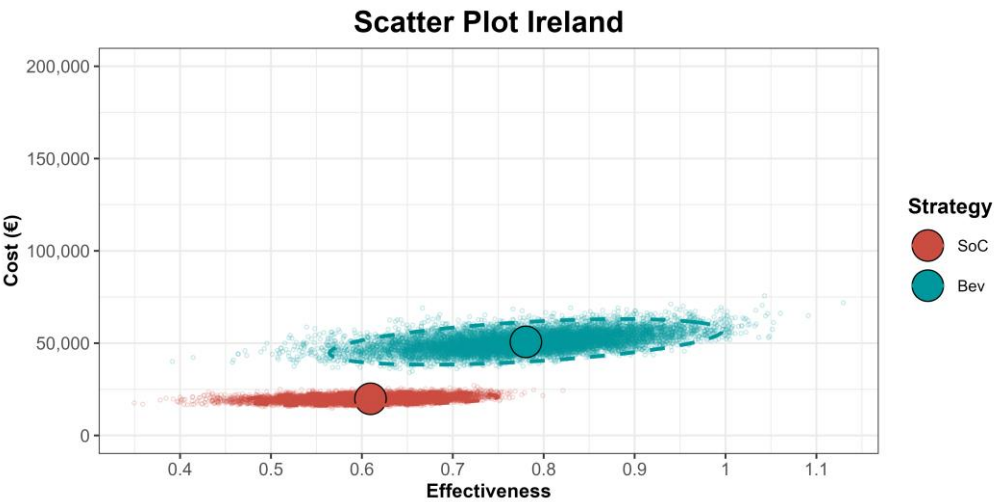
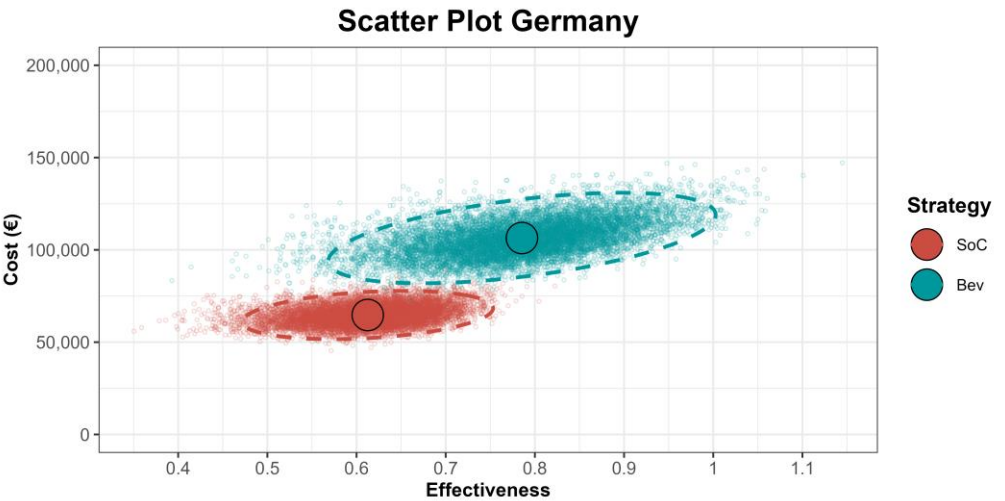


Figure 3: Cost-effectiveness Scatter Plot of Germany, Ireland and Spain, respectively.

In total, 10,000 replications were completed to determine the possible distribution of the resulting costs and effects with and without bevacizumab. The ellipses around points represent 95% confidence incidences.

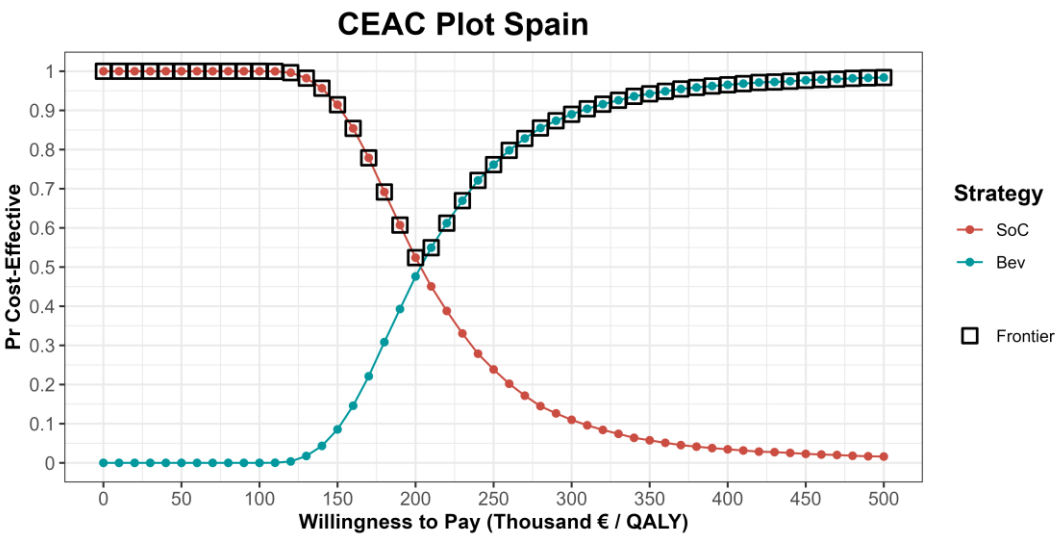
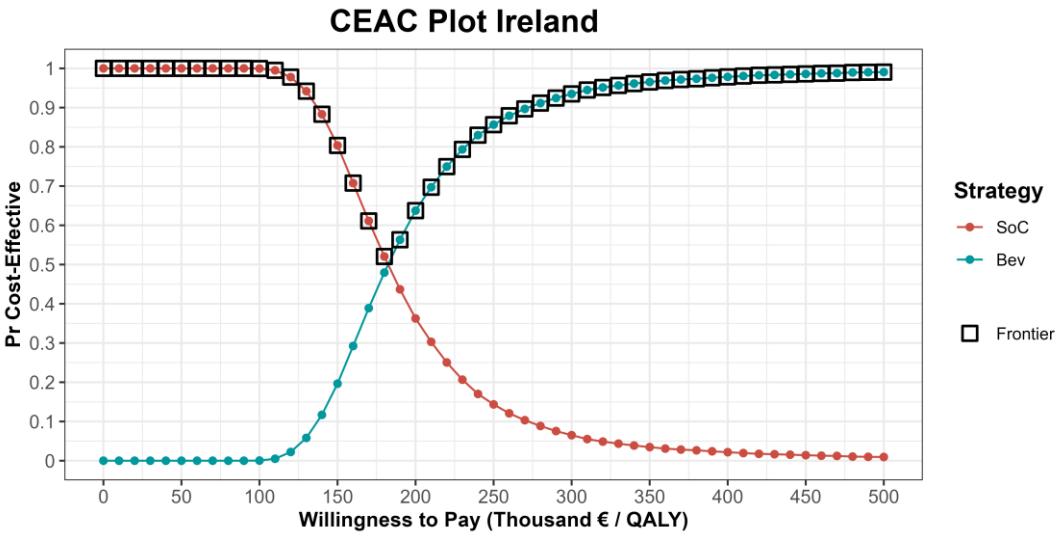
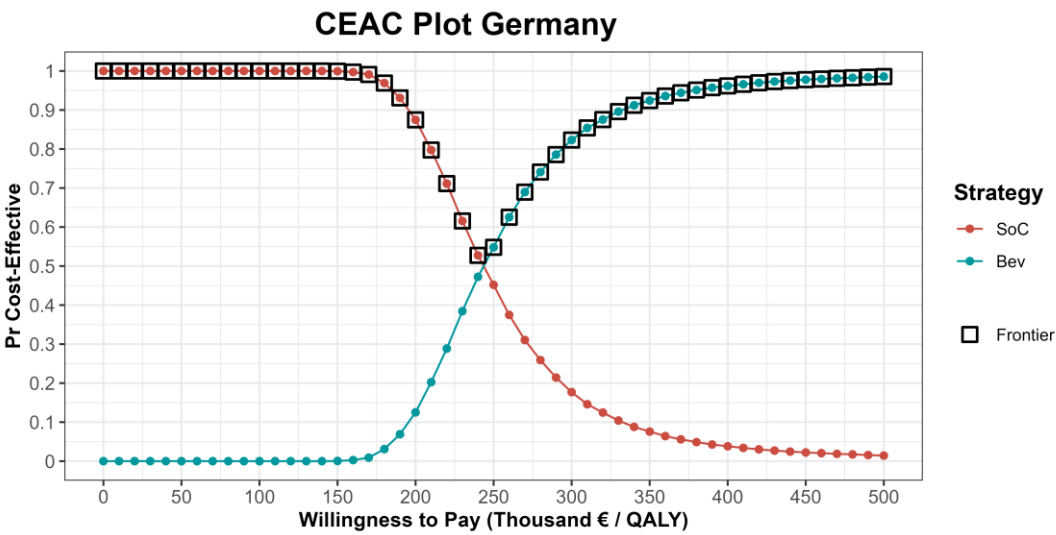


Figure 4: Cost-Effectiveness Acceptability Curves (CEACs) of Germany, Ireland and Spain, respectively.

Supplementary Information (SI)

Article title: Bevacizumab for Metastatic Colorectal Cancer with Chromosomal Instability: Cost-Effectiveness Analysis for a Novel Precision Treatment Approach.

Journal name: The European Journal of Health Economics.

Author names: Blinded.

Affiliation: Blinded.

E-mail address of the corresponding author: Blinded.

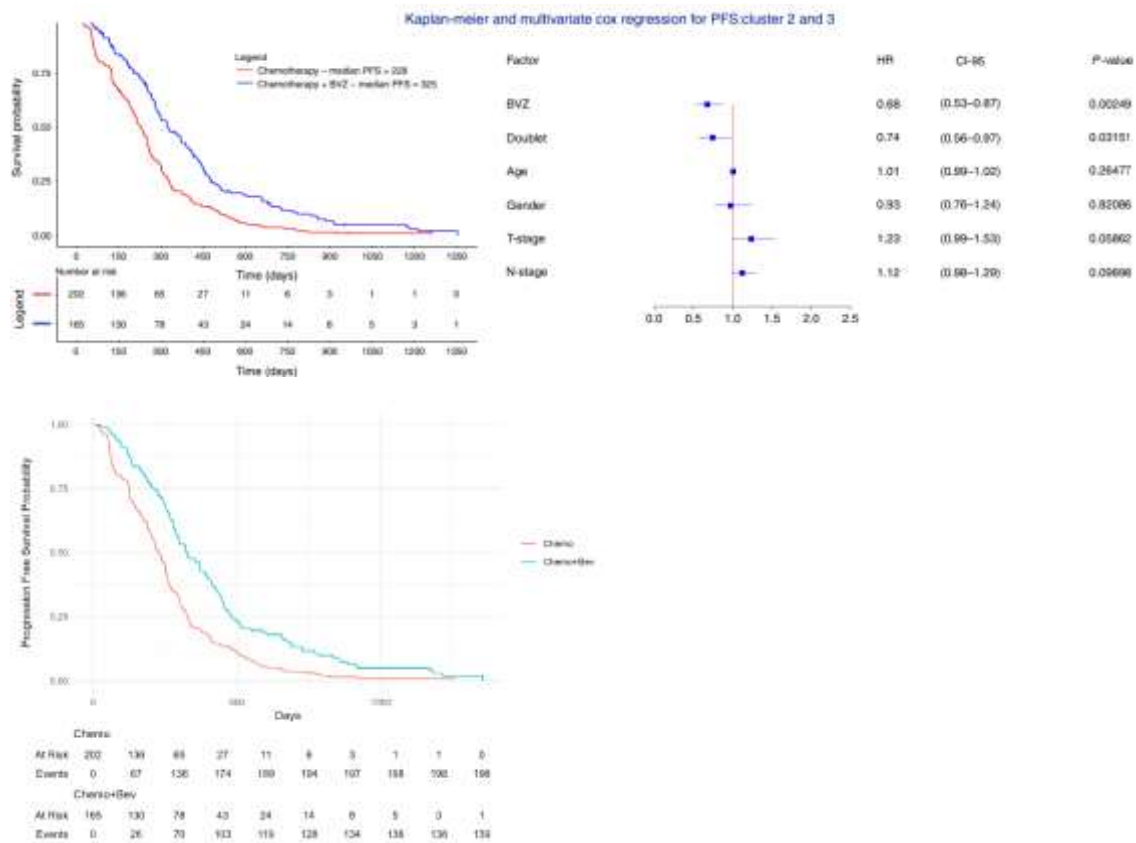


Fig. S1 PFS published in Smeets et al [19], followed by our replication of their results.

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2018;9(1):1–16. The second part shows our replication results

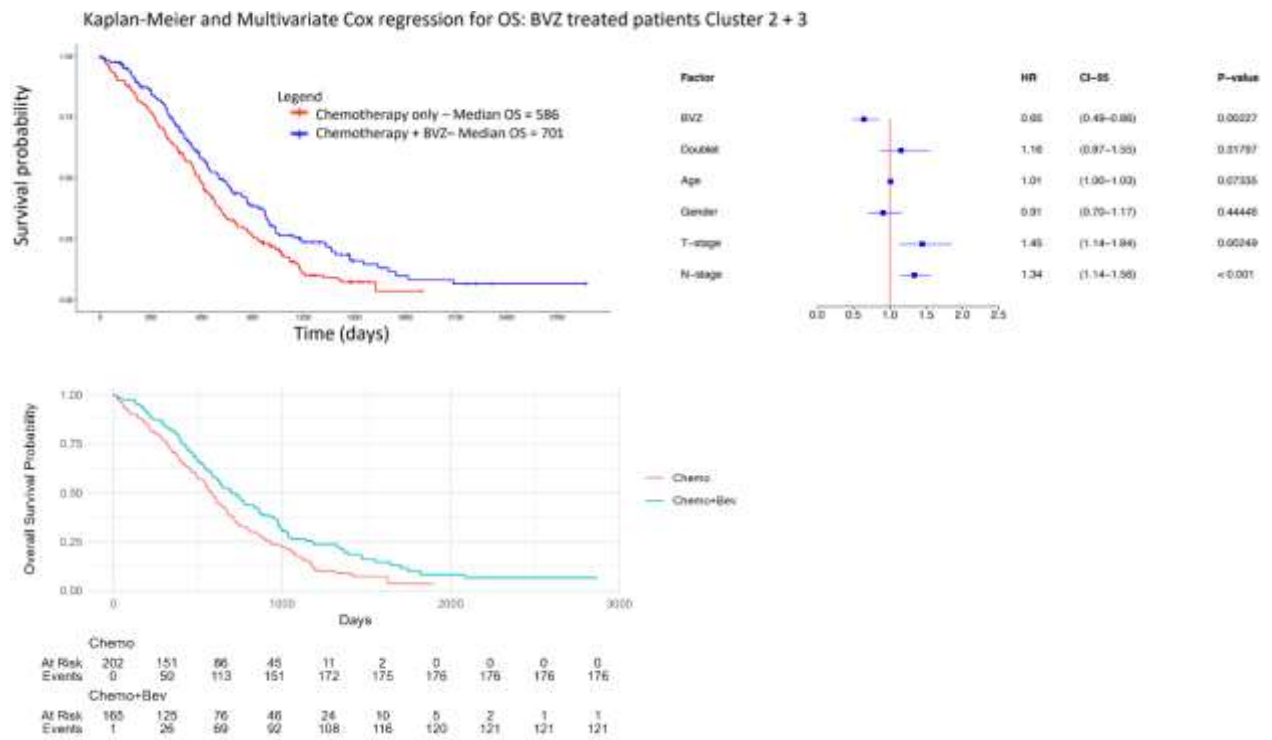


Fig. S2 OS published in Smeets et al [19], followed by our replication of their results.

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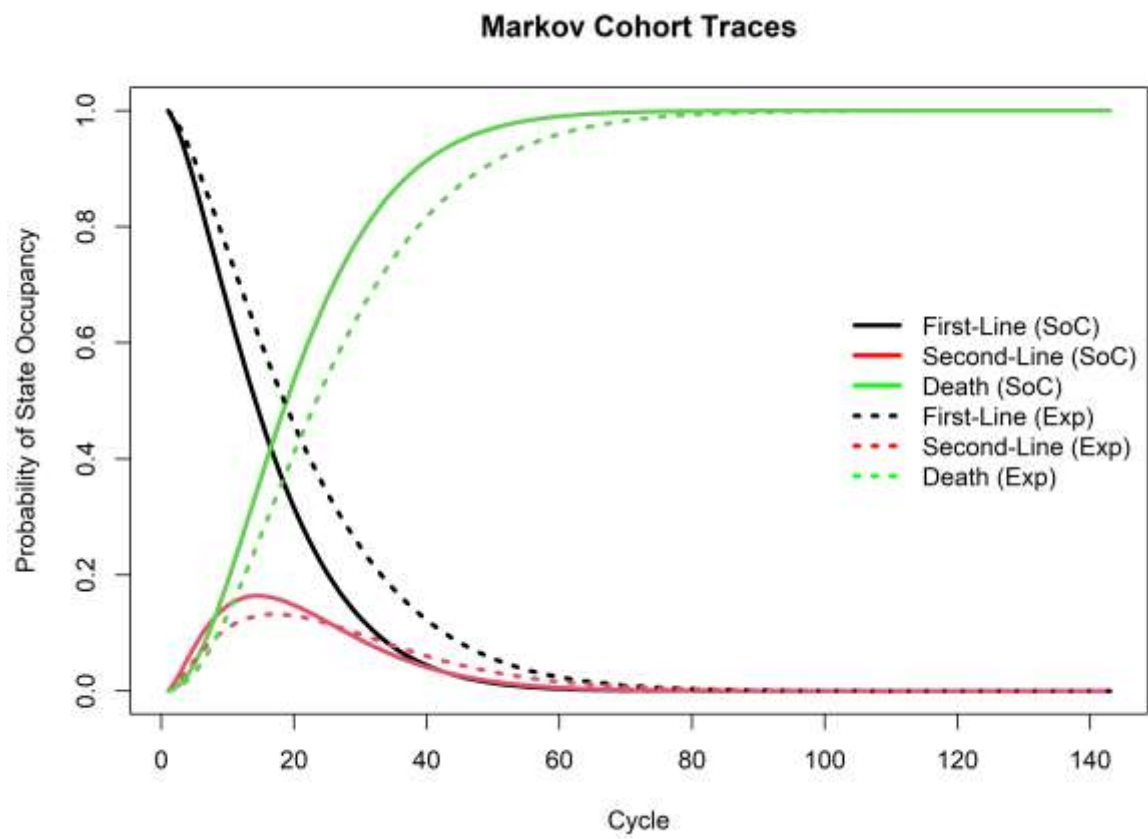


Fig. S3 Cohort Traces generated by the Markov model simulation

Distribution	Exponential	Gamma	Gompertz	Log- logistic	Log- normal	Weibull
Time-to- progression	2604.383	2570.365	2594.880	2579.625	2580.161	2574.970
Time-to- dead	2664.472	2648.915	2648.350	2665.054	2680.234	2646.087

Table S1: Akaike Information Criterion (AIC) values for the parametric models fitted to the original ANGIOPREDICT PFS and OS Kaplan-Meier survival curves.

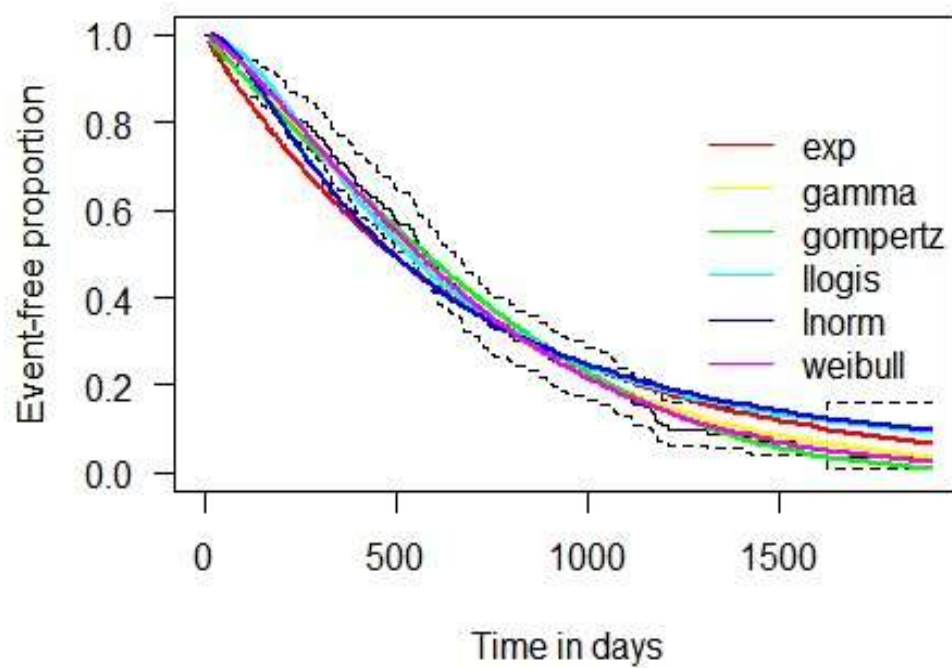
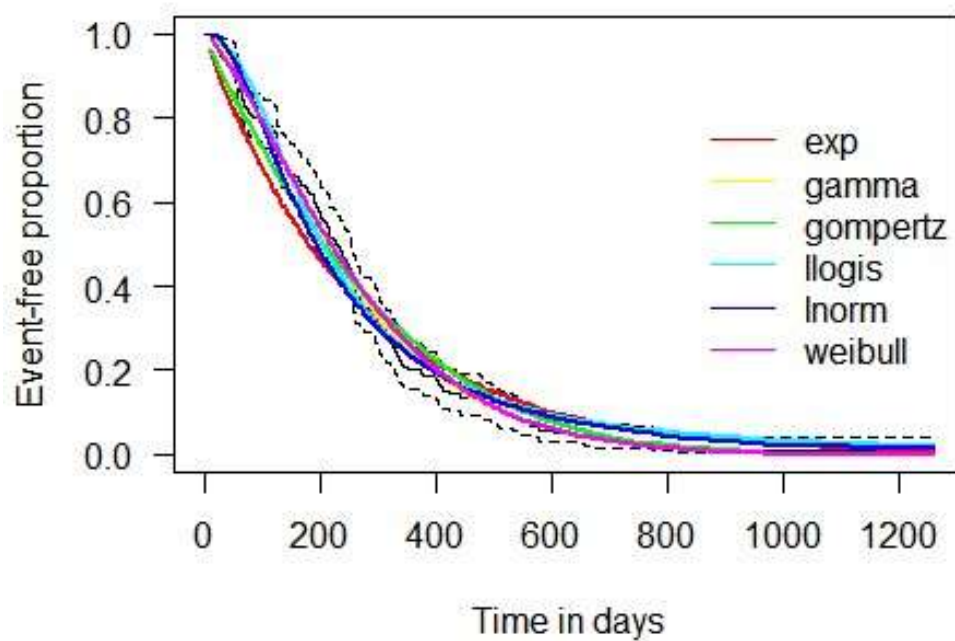


Fig. S4 Visual inspection of the relative conformity of parametric models to the original PFS and OS Kaplan-Meier survival curves, respectively

Detailed Description of Sensitivity Analysis:

In the deterministic one-way sensitivity analysis, we evaluated the influence of varying each parameter individually on the cost-effectiveness results. Assumptions for defining the interval estimates for this univariate analysis came from the literature [71]. Wherever possible, these were taken directly from the estimation process, with confidence intervals or standard errors extracted from published sources to generate a plausible range for the probabilities, adverse events, costs and utilities. Where this information was unavailable, model inputs varied 20% above and below base case values, as is standard practice [38,72]. To study the influence of a range of transition risks, the PFS and OS curves $S(t)$ in the model were altered for both treatment strategies via a hazard ratio adjustment γ , as $[S(t)^\gamma]$. Subsequently, transition risks for each cycle were calculated from these adjusted survival curves in the deterministic sensitivity analysis [38]. The adjustment hazard ratio was 1 ± 0.2 , i.e., the base case varied within plus or minus 20%. In Figure 2, we created a tornado plot of the one-way sensitivity analysis to describe which parameters are driving the majority of the variation in the outcome. In this diagram, we denote parameter levels as high or low to correspond to parameter values above or below the median point estimates reported in Table 1. This highlights where differences in the magnitude of input values (across the minimum and maximum values described in Table 1) result in changes in the expected outcome value.

While a tornado plot provides a useful visual aid to identify which individual parameters are driving the majority of the variation in the outcome of interest, varying individual parameters while fixing all others at their base case understates uncertainty, as in reality, parameters do not vary in isolation. To jointly propagate uncertainty throughout the model, we assigned distributions to any parameters not estimated with certainty. We considered whether values within a plausible range for input parameters alter the conclusions of cost-effectiveness drawn

from the original analysis. In this probabilistic sensitivity analysis (PSA), we computed model outcomes (total discounted costs and QALYs) for the experimental strategy and standard of care for each sampled set of parameter values in Table 1. Simultaneous samples were drawn from model parameters in Monte Carlo simulations according to their sampling distributions, reflecting any uncertainty in parameter estimates in the evidence. Thus, joint uncertainty was propagated through the model over 10,000 replications, providing a comprehensive assessment of model parameter uncertainty. The distribution for parameters was specified in the probabilistic analysis per standard statistical methods. Beta distributions were applied to binomial data; Gamma distributions were applied to right-skewed parameters; the log-normal distribution was applied to hazard ratios, and a multivariate normal distribution was applied to the Weibull parametric survival distribution, with all distributions applied directly in the PSA [71]. We illustrate the distribution of costs and effects for treatment with and without bevacizumab in Figure 3. We randomly sampled from each specified parameter distribution in the probabilistic analysis, calculating the expected costs and effects of that combination of parameter values. Each point in this plot represents a single sample of parameter values. Finally, we derived cost-effectiveness acceptability curves (CEACs) from the probabilistic sensitivity analysis, to describe the probability that the addition of bevacizumab is cost-effective across increasing WTP values.

Cost Calculation for the Subtyping Test:

Defining a binomial distribution with the following proportions:

cluster 1 = 0.1

cluster 2 and 3 = 0.9

The probability of finding no patients with the study subtype is:

$$P(X = 0) = 0.1^n$$

Consequently, the probability of finding at least one patient with this subtype can be expressed as:

$$P(X \geq 1) = 1 - P(X = 0)$$

$$P(X \geq 1) = 1 - 0.1^n$$

Assuming a probability of at least 95% for $P(x \geq 1)$, then n becomes:

$$0.95 = 1 - 0.1^n$$

$$n = \log(0.05) / \log(0.1) = 1.3$$

With the above assumptions a sample size of at least 2 patients would be necessary to have 95% confidence in finding at least one patient with the study subtype.

Thus, for a test costing €200 per patient, where two people need to receive the test to find one person who is intermediate to high chromosomal instability, the cost of identifying one positive patient is €200 * 2 = €400. That is, subtyping incurs a one-off cost of €400 to each patient in the experimental arm to reflect the testing cost.

Log Transformations of Hazard Ratios:

The value of ratio intervention effects have a lowest possible value of 0, a value of 1 corresponding to no intervention effect, and a highest value possible value of infinity. Logically, a ratio that halves effects and one that doubles effects should average to no effect. However, the average of 0.5 and 2 is 1.25, rather than 1. Thus, a log transformation is applied to make the hazard ratio number scale symmetric: the log of 0 is minus infinity, the log of 1 is zero, and the log of infinity is infinity. The log of 0.5 is -0.69 and the log of the OR of 2 is 0.69, such that when the halving and doubling ratios are averaged, a value of 0 is obtained, i.e., the log transformed value of 1, or no effect.

Updating Model Costs to Contemporary Values:

Model costs were inflated to 2024 values using historical Consumer Price Index (CPI) values reported by the Organisation for Economic Co-operation and Development (OECD) per country as follows:

*CPI in current year of interest / CPI in year costs were originally reported * EUR value of cost in the year the cost was reported = EUR value of the cost in the current year of interest.*